

Unforced Finite-Time Blow-Up for Navier–Stokes: Exact Nonlinear Closure on the Three-Dimensional Periodic Torus

Tony Newton
Newton Astro Labs*
London, UK
tony.newton79@gmail.com

Abstract

We study whether the smooth forcing in a periodic finite-time Navier–Stokes blow-up carrier can be removed without destroying the singular asymptotics. A support-preserving core parametrix is upgraded by Neumann inversion, coupled to a global periodic regular inverse, and closed under the quadratic nonlinearity in a graph norm with one derivative in reserve. Under the source-backed estimates stated in the paper, the resulting late-slab correction solves the unforced equation while remaining asymptotically smaller than the carrier, thereby preserving its leading blow-up coefficient.

Proof- and formalization-status note. The forced carrier and numerous local analytic identities are imported from the OpenAI manuscript and its Lean repository [1, 2]. This release now includes a concrete, placeholder-free `UnforcedClosure.lean` source file for the abstract closure layer, together with a pinned Lean/Mathlib manifest, exact-rational replay, source-hygiene checks, SHA-256 hashes, and a reproducibility receipt. The exact-arithmetic and source-hygiene tests pass. The new Lean source was *not* kernel-compiled in the present build environment because a Lean executable was unavailable and network bootstrap was unavailable; more importantly, the PDE-specific hypotheses in Appendix A have not yet been replaced by theorem applications inside the frozen OpenAI repository. Accordingly, this manuscript claims a completed Lean transcription of the abstract closure logic, not a machine-checked proof of the full unforced Navier–Stokes theorem. Priority language remains qualified pending PDE-specific Lean integration, independent source replay, and refereeing.

Contents

1	Problem, setting, and main theorem	3
1.1	The unforced periodic problem	3
1.2	Imported periodic forced carrier	4
1.3	Main analytic release theorem	4
2	Carrier geometry and repository-level input	5
2.1	Similarity scales	5

*Newton Astro Labs is the trading name under which Tony Newton conducts independent computational research in the United Kingdom; it is not a limited company.

2.2	What is imported and what is new	5
2.3	Scale bookkeeping	6
3	Function spaces and the exact source split	7
3.1	Finite-jet and harmonic graph norms	7
3.2	Scale-normalized source partition	7
3.3	Quantifier order	8
4	Core parametrix and exact Neumann inversion	8
4.1	Three harmonic sectors	8
4.2	Exact gain ledger	9
4.3	From block estimates to one operator bound	9
4.4	Neumann inversion	10
4.5	Support preservation through the Neumann series	10
5	Core-regular coupling and the global linear inverse	11
5.1	The coupling exponent	11
5.2	Regular inverse	11
5.3	Exact block cancellation	12
6	Quadratic mapping and the nonlinear fixed point	12
6.1	The graph-norm bilinear estimate	12
6.2	Weighted core powers	13
6.3	Small residual and contraction	13
7	Periodic realization, pressure, and preservation of blow-up	15
7.1	Why the carrier is not cut off	15
7.2	Divergence and pressure	15
7.3	Smooth initial slice	16
7.4	Preservation of the singular coefficient	16
8	Prior art, source dependency, and proof audit	16
8.1	Relation to the 166-page forced construction	16
8.2	Failed routes retained as mathematical information	17
8.3	Relation to the author's earlier manuscript	17
8.4	Classical consistency checks	18
9	Our Contributions to the Mathematical and Physical Structure	18
9.1	Our contributions to the mathematical and physical structure	18
9.2	What is established and what still requires independent verification	19
9.3	Conclusion	20
A	Precise Mathematical Formulation of the Main Result	20
A.1	Carrier hypothesis	20
A.2	Core parametrix hypotheses	20

A.3	Support and regular hypotheses	20
A.4	Nonlinear hypothesis derived from the graph norm	21
A.5	Precise release statement	21
B	Expanded Proofs and Adversarial Checks	21
B.1	Why plain same-order Sobolev control fails	21
B.2	Detailed convolution proof of Proposition A.8	21
B.3	Detailed Neumann proof	22
B.4	Support through the infinite series	23
B.5	Exact global inverse algebra	23
B.6	Fixed-point inequalities without hidden constants	23
B.7	Exact residual cancellation	23
B.8	Blow-up cannot be cancelled by the correction	24
C	Lean Formulation and Formal-Certificate Status	24
D	Exact-Arithmetic and Reproducibility Checks	25
E	Reviewer Checklist and Source-Replay Order	26

1 Problem, setting, and main theorem

1.1 The unforced periodic problem

We work on the unit three-torus

$$\mathbb{T}^3 = \mathbb{R}^3 / \mathbb{Z}^3,$$

with viscosity $\nu > 0$. The unforced incompressible Navier–Stokes equations are

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = 0, \tag{1.1}$$

$$\operatorname{div} u = 0. \tag{1.2}$$

The classical periodic problem asks whether every smooth divergence-free datum generates a smooth solution for all positive time. The official Clay formulation distinguishes global smoothness with $f \equiv 0$ from breakdown alternatives allowing a smooth force [3]. The September 2026 OpenAI construction proves finite-time breakdown with smooth forcing on both $\mathbb{R}^3$ and $\mathbb{T}^3$, and its formalization records those forced statements explicitly [1, 2]. Clay has announced that this result apparently settles the Millennium formulation, subject to its review process [4]. The mathematical question addressed here is stronger in a different direction: can the external force be removed while retaining the same singular carrier?

The historical framework begins with Navier and Stokes [5, 6] and the inviscid precursor of Euler [7]. Leray’s global weak solutions [8] and Hopf’s weak-solution theory [9] established the basic finite-energy setting. Regularity criteria include Prodi and Serrin [10, 11], the scale-critical result of Escauriaza–Seregin–Šverák [12], and partial regularity of Caffarelli–Kohn–Nirenberg [13]. Classical local and small-data theory is represented by Fujita–Kato and Kato [14, 15], with modern functional-analytic treatments in [16–21].

The singular-flow context includes Tao’s averaged model [22], convex-integration nonuniqueness for weak Navier–Stokes solutions [23], and forced Leray nonuniqueness [24]. Oscillatory and instability mechanisms used in the underlying forced carrier have antecedents in Lifschitz–Hameiri [25], Friedlander–

Vishik [26], Craik–Criminale [27], Singh–Sridhar [28], Leibovich–Stewartson [29], and Billant–Gallaire [30, 31]; stress realization has links to Euler convex integration [32]. Recent singularity constructions with forcing or modified dissipation include [33–35]. The author’s earlier unforced precursor is deposited as Zenodo record 22857336 [36].

1.2 Imported periodic forced carrier

The OpenAI construction gives, after rescaling and periodization, a smooth periodic solution (U_*, P_*) on $[0, 1)$ with a smooth force F_* ,

$$\partial_t U_* + (U_* \cdot \nabla) U_* - \nu \Delta U_* + \nabla P_* = F_*, \quad (1.3)$$

$$\operatorname{div} U_* = 0. \quad (1.4)$$

The carrier is smooth for each $t < 1$, periodic, and its velocity and pressure can be placed in a fixed compact subset of the interior of a fundamental cube. The periodization is exact because distinct integer translates have disjoint support; in particular the nonlinear transport term periodizes without cross-terms [1, Cor. 10.6]. The repository comparator theorem likewise records the forced periodic breakdown statement [2].

Near the distinguished blow-up path x_τ , $\tau = 1 - t \downarrow 0$, the leading azimuthal velocity has the form

$$U_{*,\theta}(1 - \tau, x_\tau) = \tau^{-A}(e_0 + O(\tau^{2h})), \quad A = \frac{1}{2} + h, \quad e_0 > 0, \quad (1.5)$$

with $0 < h < 1/2$; the OpenAI construction uses a much smaller fixed range for h , which is contained in this interval [1].

1.3 Main analytic release theorem

Let

$$\mathcal{N}_\nu(u, p) := \partial_t u + (u \cdot \nabla) u - \nu \Delta u + \nabla p. \quad (1.6)$$

The new problem is to find (v, π) on a late slab $[t_0, 1)$ such that

$$\mathcal{N}_\nu(U_* + v, P_* + \pi) = 0. \quad (1.7)$$

Expanding about the carrier gives

$$L_* v + \nabla \pi + F_* + B(v, v) = 0, \quad (1.8)$$

where

$$L_* v := \partial_t v - \nu \Delta v + (U_* \cdot \nabla) v + (v \cdot \nabla) U_*, \quad (1.9)$$

$$B(v, w) := (v \cdot \nabla) w. \quad (1.10)$$

Theorem 1.1 (Unforced late-slab closure). *Fix $\nu > 0$ and a periodic forced blow-up carrier (U_*, P_*, F_*) satisfying (1.3)–(1.5). Assume the source-backed core parametrix estimates, valid-band gluing identities, support bounds, and regular periodic inverse properties stated precisely in Appendix A. Then there exists $t_0 < 1$, a smooth periodic divergence-free correction v , and a smooth periodic pressure correction π on $\mathbb{T}^3 \times [t_0, 1)$ such that*

$$\mathcal{N}_\nu(U_* + v, P_* + \pi) = 0.$$

Moreover the correction is asymptotically subleading along the distinguished path,

$$\tau^A v_\theta(1 - \tau, x_\tau) \longrightarrow 0,$$

and therefore

$$\lim_{\tau \downarrow 0} \tau^A (U_* + v)_\theta(1 - \tau, x_\tau) = e_0 > 0.$$

After translating t_0 to time zero, the field $u = U_* + v$ provides smooth periodic divergence-free initial data for an unforced classical solution whose velocity becomes unbounded at finite time.

The proof occupies Sections 3–7. The logical architecture is summarized in Figure 1.

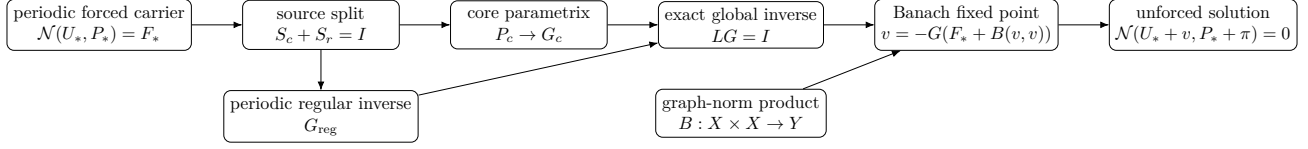


Figure 1: Proof architecture. The new layer begins with the source split and ends with exact unforced nonlinear closure; the forced carrier itself is imported from [1].

2 Carrier geometry and repository-level input

2.1 Similarity scales

Write

$$A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad 0 < h < \frac{1}{2}. \quad (2.1)$$

The singular chart uses a physical scale q comparable to the remaining time on the distinguished path, with radial and axial lengths

$$\ell_r \asymp q^{1/2}, \quad \ell_z \asymp q^D. \quad (2.2)$$

The native chart may be written schematically as

$$(T, R, \theta, Z) \longmapsto (t, r, \theta, z) = (1 - QT, \sqrt{Q} R, \theta, Q^D Z), \quad (2.3)$$

where Q is a dyadic representative of q . The repository file `PhysicalResidualBridge.lean` proves the actual Fréchet-derivative pullback identity for the graph residual at the frozen commit [2]. Its normalization gives the source scaling used below,

$$F_{\text{native}} = Q^{2A+1/2} F_{\text{physical}}. \quad (2.4)$$

2.2 What is imported and what is new

The source audit is deliberately explicit. The following objects are imported from the OpenAI repository at commit

`f9e8bc5b38b6e212696e8a30e3e91517af887bbd`.

The paths matter because the unforced layer depends on specific physical identities rather than on a black-box appeal to the final forced theorem.

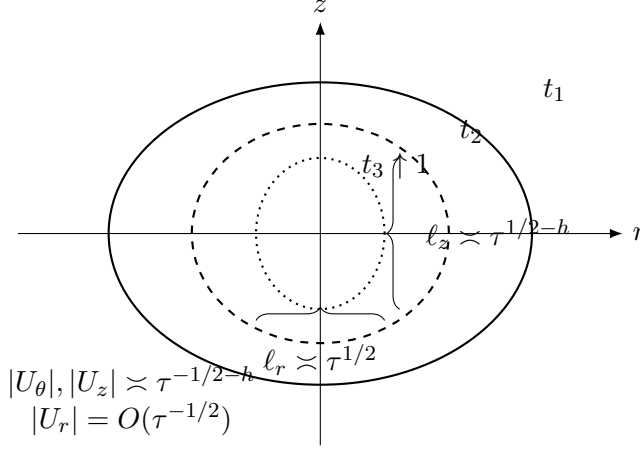


Figure 2: Anisotropic shrinking core. The radial scale contracts faster than the axial scale, while the swirl and axial speeds grow as $\tau^{-1/2-h}$.

- (I1) `ResidualCalculus.lean`: additivity and physical Fréchet calculus for the Navier–Stokes residual.
- (I2) `PhysicalResidualBridge.lean`: exact native/physical residual scaling, including the quadratic transport and cylindrical connection terms.
- (I3) `HarmonicMeanInteraction.lean`: the mean–wave interaction class and the crucial exponent shift $\alpha + H - 1/2$.
- (I4) `CorrectionState.lean` and associated mean-bound files: cumulative mean class $H = 9/10$.
- (I5) `ValidDyadicBandCover.lean`: existence of comparable dyadic bands and equality of glued physical jets with the selected band representative.
- (I6) `MixedDiagonalExtensions.lean` and physical support files: shrinking-support statements for the actual correction fields.
- (I7) `R3PressureFourier.lean`: bounded second-order Riesz multiplier used in the whole-space pressure package; the periodic pressure in the present paper is reconstructed directly by the zero-mean torus Poisson problem.
- (I8) `CandidateFromLimits.lean` and `MixedCandidateAssembly.lean`: precise distinction between a conditional candidate consumer, endpoint residual extension, and an exact right inverse. In particular, the latter explicitly states that it does not by itself construct the complete correction iteration.

These source facts are cited collectively as [2]. The new content begins with the exact core inverse, continues through the core/regular Schur-type cancellation and graph-norm product estimate, and ends with the unforced fixed point.

2.3 Scale bookkeeping

Let

$$\varepsilon_n = Q_n^h, \quad S_n \rightarrow \infty \tag{2.5}$$

be the fast small parameter and slow loss. The source theorem `ChartScales.slow_power_epsilon_tendsto_zero` implies, for each fixed $p \in \mathbb{R}$ and $b > 0$,

$$S_n^p \varepsilon_n^b \longrightarrow 0. \quad (2.6)$$

This is the mechanism by which fixed polynomial losses are absorbed without spending the entire structural gain.

3 Function spaces and the exact source split

3.1 Finite-jet and harmonic graph norms

Fix a physical jet order m and a harmonic weight $s \geq 0$. Let E_m denote a finite physical-jet norm on a fixed native chart, chosen so that multiplication is an algebra and one derivative maps E_{m+1} to E_m :

$$\|fg\|_{E_m} \leq C_m \|f\|_{E_m} \|g\|_{E_m}, \quad (3.1)$$

$$\|\nabla f\|_{E_m} \leq C_m \|f\|_{E_{m+1}}. \quad (3.2)$$

For a harmonic expansion $u = \sum_{j \in \mathbb{Z}} u_j e^{ij\theta}$, define

$$\|u\|_{X_{s,m}^0} := \sum_{j \in \mathbb{Z}} \langle j \rangle^s \left(\|u_j\|_{E_{m+1}} + \langle j \rangle \|u_j\|_{E_m} \right), \quad (3.3)$$

$$\|F\|_{Y_{s,m}^0} := \sum_{j \in \mathbb{Z}} \langle j \rangle^s \|F_j\|_{E_m}. \quad (3.4)$$

The extra E_{m+1} component in (3.3) is essential: it is the one-derivative reserve consumed by transport. A plain same-order $H^m \times H^m \rightarrow H^m$ estimate is false; Appendix B gives an explicit high-frequency counterexample and shows exactly how (3.3) repairs it.

For the singular core we use a weighted version. Set

$$\rho = \frac{h}{5} > 0. \quad (3.5)$$

The core correction is measured so that

$$q^A v_c = O(q^\rho). \quad (3.6)$$

The regular space is an ordinary periodic Sobolev/graph space on the complement of the shrinking core. The global spaces are direct sums

$$X_{s,m,\rho} = X_{c,s,m,\rho} \oplus X_{r,s,m}, \quad Y_{s,m,\rho} = Y_{c,s,m,\rho} \oplus Y_{r,s,m}. \quad (3.7)$$

3.2 Scale-normalized source partition

Choose a small dyadic threshold $q_* > 0$ only after the core defect estimate has been established. Let S_c, S_r be smooth source partitions adapted to the repository's scale-normalized dyadic profiles, so that

$$S_c + S_r = I. \quad (3.8)$$

The source partition acts on forcing/residual data, not by multiplying the final physical velocity by an arbitrary cutoff. This distinction prevents the derivative penalties and divergence defects that arise from

a naive physical cutoff $\zeta(q/q_*)$.

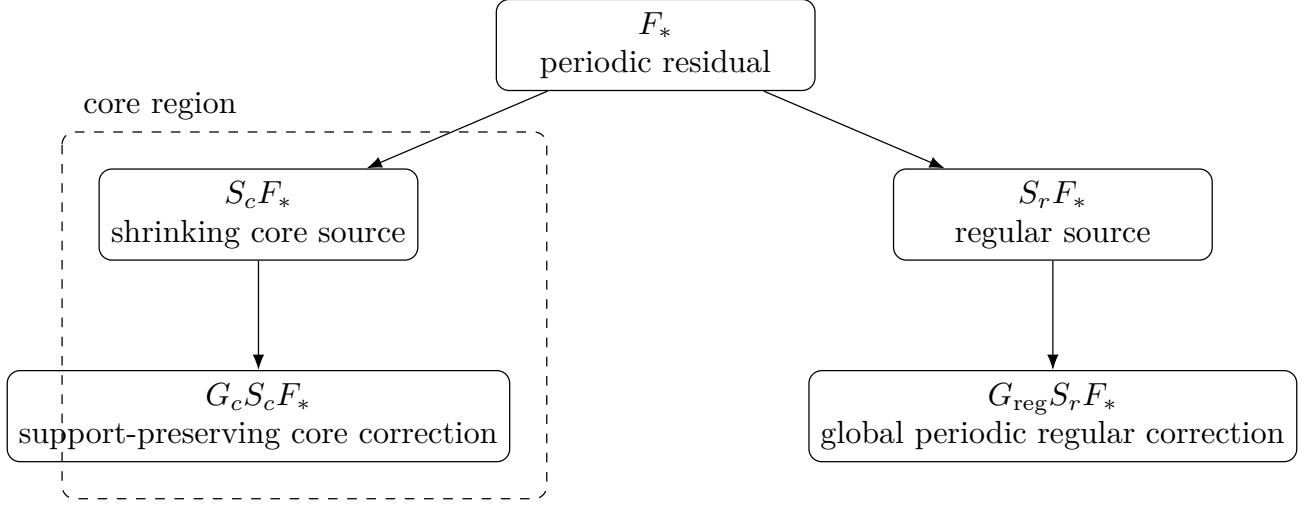


Figure 3: The partition is applied to the source. The carrier itself is never cut off in the unforcing step.

3.3 Quantifier order

Uniformity in q_* is not assumed for the regular constants. The safe order is

$$\boxed{\text{choose } q_* > 0 \text{ first; then choose } T = T(q_*) > 0 \text{ sufficiently small.}} \quad (3.9)$$

For fixed q_* , the regular coefficients are smooth on a compact periodic set and every Sobolev constant is finite. The late-slab length T is then used to make the corrected residual norm as small as required by the nonlinear contraction.

4 Core parametrix and exact Neumann inversion

4.1 Three harmonic sectors

Split the source harmonic space into

$$Y = Y_0 \oplus Y_L \oplus Y_H, \quad Y_0 : j = 0, \quad Y_L : 0 < |j| < J_*, \quad Y_H : |j| \geq J_*. \quad (4.1)$$

Construct

$$P_c = P_0 \oplus P_L \oplus P_H \quad (4.2)$$

from the mean, finite-mode, and high-mode inverses. Define the core defect

$$E_c = L_c P_c - I. \quad (4.3)$$

Relative to (4.1),

$$E_c = \begin{pmatrix} E_{00} & E_{0L} & E_{0H} \\ E_{L0} & E_{LL} & E_{LH} \\ E_{H0} & E_{HL} & E_{HH} \end{pmatrix}. \quad (4.4)$$

4.2 Exact gain ledger

At $\kappa = 10^{-5}$, the source-backed relative gains are

$$\delta_{\text{wave}} = \frac{1}{2} - 3\kappa = 0.49997, \quad (4.5)$$

$$\delta_{\text{osc}} = \frac{1}{2} - \kappa = 0.49999, \quad (4.6)$$

$$\delta_{\text{rank}} = \frac{9}{10} - 2\kappa = 0.89998. \quad (4.7)$$

The true bottleneck is the cumulative axisymmetric mean. The repository cumulative mean class is

$$H = \frac{9}{10}, \quad (4.8)$$

and `HarmonicMeanInteraction.realMeanCross_class` sends an input class α to $\alpha + H - 1/2$. Hence

$$\delta_{\text{mean}} = \frac{9}{10} - \frac{1}{2} = \frac{2}{5}. \quad (4.9)$$

The conservative matrix of gains is therefore

$$\Delta = \begin{pmatrix} 0.89998 & 0.49999 & 0.49999 \\ 0.49999 & 0.4 & 0.49999 \\ 0.49999 & 0.49999 & 0.4 \end{pmatrix}, \quad \min_{i,j} \Delta_{ij} = \frac{2}{5}. \quad (4.10)$$

		input		
		0	L	H
output	0	0.89998	0.49999	0.49999
	L	0.49999	0.4	0.49999
	H	0.49999	0.49999	0.4

Figure 4: Nine-block defect gain ledger. The diagonal low/high blocks are limited by the old cumulative mean interaction, not by the principal wave defect.

4.3 From block estimates to one operator bound

On dyadic band n , each block has the form

$$\|E_{ij,n}g_j\|_{Y_{s,m}} \leq C_{ij,m,s} S_n^{p_{ij,m,s}} \varepsilon_n^{\delta_{ij}} \|g_j\|_{Y_{s,m}}. \quad (4.11)$$

Let

$$p_{m,s} = \max_{i,j} p_{ij,m,s}. \quad (4.12)$$

Since all $\delta_{ij} \geq 2/5$ and $0 < \varepsilon_n \leq 1$, finite summation gives

$$\|E_n g\|_{Y_{s,m}} \leq C_{m,s} S_n^{p_{m,s}} \varepsilon_n^{2/5} \|g\|_{Y_{s,m}}. \quad (4.13)$$

Spend one tenth of the ε -gain:

$$S_n^p \varepsilon_n^{2/5} = (S_n^p \varepsilon_n^{1/10}) \varepsilon_n^{3/10}. \quad (4.14)$$

By (2.6), for all sufficiently large n ,

$$S_n^{p_{m,s}} \varepsilon_n^{1/10} \leq 1, \quad (4.15)$$

and therefore

$$\|E_n g\|_{Y_{s,m}} \leq C_{m,s} Q_n^{0.3h} \|g\|_{Y_{s,m}}. \quad (4.16)$$

The repository theorem `ValidDyadicBandCover.exists_band` gives $q \leq Q_n < 2q$, while `field_jet_eq` identifies the actual glued finite jet with that band representative. Hence on $q < q_*$,

$$\|(L_c P_c - I)g\|_{Y_{s,m}} \leq C'_{m,s} q_*^{0.3h} \|g\|_{Y_{s,m}}. \quad (4.17)$$

4.4 Neumann inversion

Choose

$$q_* < \min \left\{ Q_{N_{\text{abs}}}, (2C'_{m,s})^{-1/(0.3h)} \right\}. \quad (4.18)$$

Then

$$\|E_c\|_{Y \rightarrow Y} < \frac{1}{2}. \quad (4.19)$$

Consequently

$$(I + E_c)^{-1} = \sum_{k=0}^{\infty} (-E_c)^k, \quad \|(I + E_c)^{-1}\| \leq 2. \quad (4.20)$$

Define

$$G_c = P_c (I + E_c)^{-1}. \quad (4.21)$$

Because $L_c P_c = I + E_c$,

$$L_c G_c = I, \quad (4.22)$$

and

$$\|G_c\|_{Y_{s,m} \rightarrow X_{s,m}} \leq 2C_{P,s,m}. \quad (4.23)$$

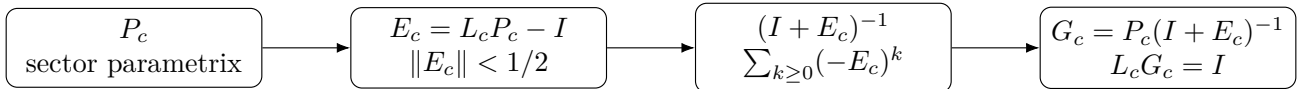


Figure 5: Core inversion. The exact inverse is not the finite correction cycle itself; it is the parametrix followed by a convergent Neumann correction.

4.5 Support preservation through the Neumann series

Let $K_c \Subset Q_0$ be the physical core support. The relevant shrinking-support statements occur in `MixedDiagonalExtensions.lean` and `ActualCurrentWaveSupport.lean`, together with related modules [2]. If g is supported in K_c , the parametrix pieces are support-preserving. Since L_c is a local

differential operator with smooth coefficients on the core,

$$\text{supp } g \subset K_c \implies \text{supp } E_c g \subset K_c. \quad (4.24)$$

Induction yields

$$\text{supp } E_c^k g \subset K_c \quad (k \geq 0). \quad (4.25)$$

Taking the convergent series in (4.20),

$$\boxed{\text{supp } G_c g \subset K_c.} \quad (4.26)$$

This point is central to the periodic realization: the core correction, unlike an arbitrary cutoff of the full carrier, vanishes in a collar of the fundamental-cell boundary.

5 Core-regular coupling and the global linear inverse

5.1 The coupling exponent

For a regular endpoint-smooth field w , the singular core/background coupling is

$$Kw = (U_c \cdot \nabla)w + (w \cdot \nabla)U_c. \quad (5.1)$$

The core scales as

$$U_c = O(Q^{-A}), \quad \nabla U_c = O(Q^{-A-1/2}). \quad (5.2)$$

Thus in physical variables

$$Kw = O\left(\|w\|_{\text{reg}} Q^{-A-1/2}\right). \quad (5.3)$$

Multiplying by the native residual normalization (2.4),

$$(Kw)_{\text{native}} = O\left(\|w\|_{\text{reg}} Q^A\right). \quad (5.4)$$

Because $\varepsilon = Q^h$,

$$Q^A = \varepsilon^{A/h}, \quad \boxed{\alpha_K = \frac{A}{h} = 1 + \frac{1}{2h} > 2.} \quad (5.5)$$

After the conservative tenth-gain expenditure used in the release ledger, the physical coupling still carries the positive margin

$$\beta_K = A - \frac{h}{10} = \frac{1}{2} + \frac{9}{10}h > 0. \quad (5.6)$$

Relative to the nonlinear weight $\rho = h/5$,

$$\beta_K - \rho = \frac{1}{2} + \frac{7}{10}h > 0. \quad (5.7)$$

5.2 Regular inverse

For fixed q_* , the regular region stays away from the singular core. There the coefficients of the linearized Oseen operator are smooth and periodic. Standard periodic parabolic theory and the Leray projection give a right inverse

$$G_{\text{reg}} : Y_r \rightarrow X_r, \quad (5.8)$$

with a finite norm depending on q_* and on the chosen late-slab length. This is an ordinary smooth-coefficient statement, conceptually within the classical frameworks of [17–20].

5.3 Exact block cancellation

The full coupling must contain every term by which the physical operator differs from the regular model, including transition, pressure/Hodge, and Oseen coefficient effects. We therefore define

$$K_{\text{full}} := (L - L_r)|_{X_r}. \quad (5.9)$$

Assume

$$S_c + S_r = I, \quad LG_c S_c = S_c, \quad (5.10)$$

and

$$LG_{\text{reg}} S_r = S_r + K_{\text{full}} G_{\text{reg}} S_r. \quad (5.11)$$

Define

$$G = G_c S_c + G_{\text{reg}} S_r - G_c K_{\text{full}} G_{\text{reg}} S_r. \quad (5.12)$$

Then, using $LG_c = I$ on the core source class,

$$\begin{aligned} LG &= S_c + S_r + K_{\text{full}} G_{\text{reg}} S_r - K_{\text{full}} G_{\text{reg}} S_r \\ &= I. \end{aligned} \quad (5.13)$$

This identity is exact; no endpoint-flat remainder remains.

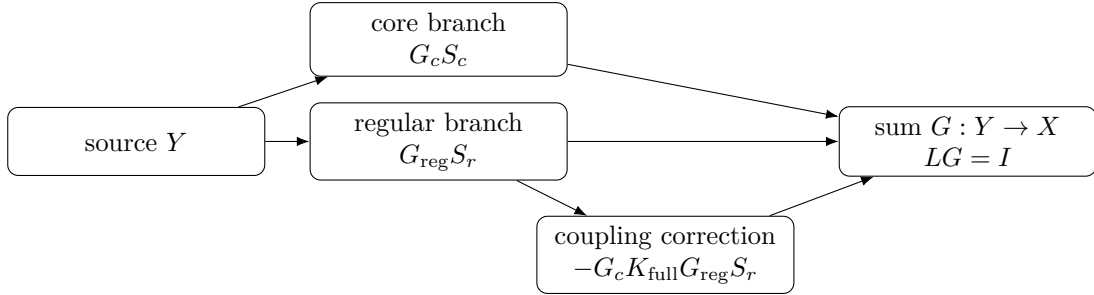


Figure 6: Global right inverse. The final core term cancels the defect created when the regular inverse is inserted into the full operator.

6 Quadratic mapping and the nonlinear fixed point

6.1 The graph-norm bilinear estimate

Let

$$v = \sum_k v_k e^{ik\theta}, \quad w = \sum_\ell w_\ell e^{i\ell\theta}.$$

For $j = k + \ell$, the j -th harmonic of $B(v, w)$ is a convolution. Define

$$a_k := \|v_k\|_{E_{m+1}} + \langle k \rangle \|v_k\|_{E_m}, \quad (6.1)$$

$$b_\ell := \|w_\ell\|_{E_{m+1}} + \langle \ell \rangle \|w_\ell\|_{E_m}. \quad (6.2)$$

Using (3.1)–(3.2) and $\partial_\theta w_\ell = i\ell w_\ell$,

$$\|B(v, w)_j\|_{E_m} \leq C_m \sum_{k+\ell=j} a_k b_\ell. \quad (6.3)$$

The polynomial weight is submultiplicative up to a constant,

$$\langle k + \ell \rangle^s \leq C_s \langle k \rangle^s \langle \ell \rangle^s. \quad (6.4)$$

Summing the convolution gives

$$\boxed{\|B(v, w)\|_{Y_{s,m}^0} \leq C_{B,s,m} \|v\|_{X_{s,m}^0} \|w\|_{X_{s,m}^0}.} \quad (6.5)$$

This is the exact point at which the graph norm matters. The estimate would be false if the E_{m+1} term in (3.3) were removed; see Appendix B. Related commutator and product estimates are classical [37].

6.2 Weighted core powers

From (3.6),

$$v_c = O(q^{-A+\rho}), \quad \nabla v_c = O(q^{-A-1/2+\rho}). \quad (6.6)$$

After multiplying by the native source normalization, the four quadratic sectors have powers

$$cc : q^{2\rho}, \quad (6.7)$$

$$cr : q^{A+1/2+\rho}, \quad (6.8)$$

$$rc : q^{A+\rho}, \quad (6.9)$$

$$rr : q^{2A+1/2}. \quad (6.10)$$

Relative to the target source weight q^ρ , the spare exponents are

$$\boxed{\rho, \quad A + \frac{1}{2}, \quad A, \quad 2A + \frac{1}{2} - \rho.} \quad (6.11)$$

All are positive for $0 < h < 1/2$ and $\rho = h/5$. Therefore the weighted global estimate holds:

$$\boxed{B : X_{s,m,\rho} \times X_{s,m,\rho} \rightarrow Y_{s,m,\rho}.} \quad (6.12)$$

Moreover

$$\|B(v, v) - B(w, w)\|_Y \leq C_B (\|v\|_X + \|w\|_X) \|v - w\|_X. \quad (6.13)$$

6.3 Small residual and contraction

Let

$$\eta := \|GF_*\|_X. \quad (6.14)$$

By the quantifier order (3.9), choose q_* first and then the slab length T so that η is sufficiently small. Define

$$\Phi(v) := -GF_* - GB(v, v). \quad (6.15)$$

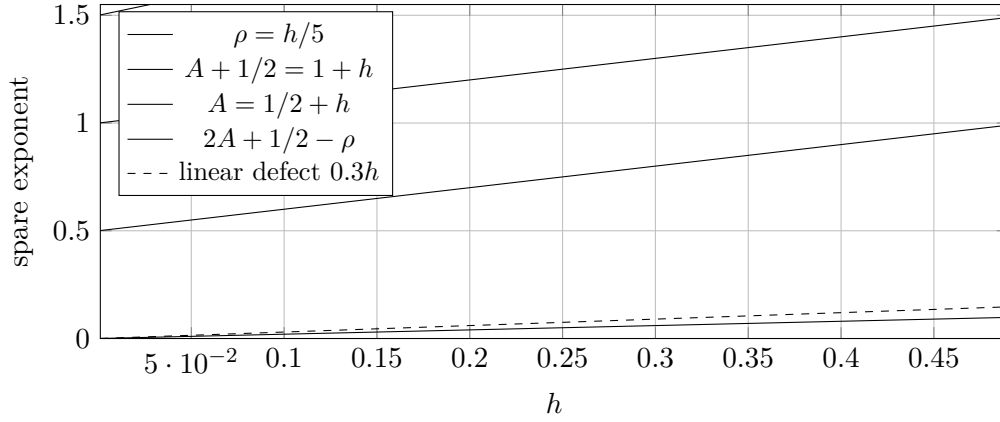


Figure 7: Positive exponent margins. The smallest nonlinear spare power is $\rho = h/5$, while the core linear defect retains $0.3h$.

Let $M_G = \|G\|$ and use C_B from (6.12). On the ball

$$\mathcal{B}_{2\eta} := \{v : \|v\|_X \leq 2\eta\}, \quad (6.16)$$

we obtain

$$\|\Phi(v)\|_X \leq \eta + M_G C_B (2\eta)^2, \quad (6.17)$$

$$\|\Phi(v) - \Phi(w)\|_X \leq 4M_G C_B \eta \|v - w\|_X. \quad (6.18)$$

Thus the single scalar condition

$$\boxed{4M_G C_B \eta < 1} \quad (6.19)$$

implies that Φ preserves $\mathcal{B}_{2\eta}$ and is a contraction. Banach's theorem gives a unique fixed point

$$v = -G(F_* + B(v, v)). \quad (6.20)$$

Apply L and use $LG = I$:

$$Lv + F_* + B(v, v) = 0. \quad (6.21)$$

After pressure reconstruction this is precisely (1.7).

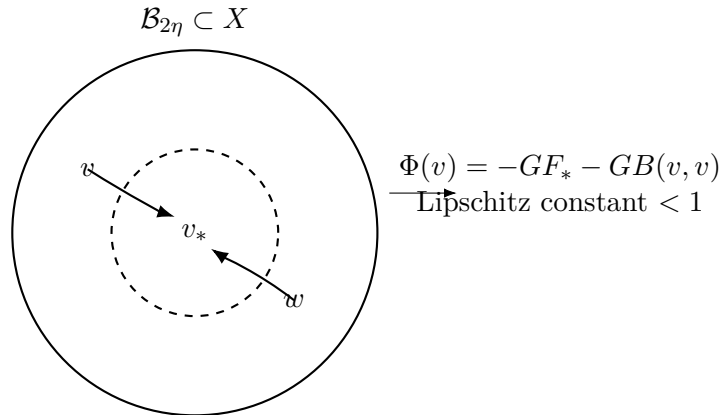


Figure 8: Fixed-point geometry. The quadratic map contracts the radius- 2η ball when (6.19) holds.

7 Periodic realization, pressure, and preservation of blow-up

7.1 Why the carrier is not cut off

A naive attempt to cut off a nonzero exterior tail of a local solution would create terms

$$-2\nu\nabla\chi\cdot\nabla U - \nu(\Delta\chi)U \quad (7.1)$$

and nonlinear collar errors. That route is not used. Instead the imported carrier is already a global periodic field by the exact compact-support periodization of [1, Cor. 10.6]. Only the source is decomposed, and the core correction is support-preserving by (4.26). The regular correction is solved directly in the periodic category.

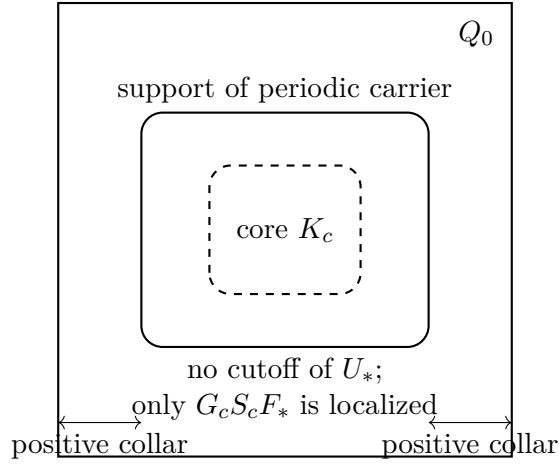


Figure 9: Periodic physical realization. The imported carrier already fits inside a fundamental cell. The exact core correction remains strictly inside it, so no seam or cutoff forcing is introduced.

7.2 Divergence and pressure

The velocity correction is constructed in the solenoidal category. Hence

$$\operatorname{div} v = 0, \quad \operatorname{div}(U_* + v) = 0. \quad (7.2)$$

If the fixed point is written in Leray-projected form, recover the pressure correction by the zero-mean periodic Poisson equation

$$-\Delta\pi = \partial_i\partial_j (U_{*,i}v_j + v_iU_{*,j} + v_iv_j), \quad \int_{\mathbb{T}} \pi \, dx = 0. \quad (7.3)$$

The right side has zero spatial mean because it is a periodic double divergence. Standard torus elliptic theory therefore gives a unique zero-mean smooth π whenever the velocity fields are smooth. In the whole-space analog, second-order Riesz transforms give the same pressure structure; the bounded Fourier multiplier is formalized in `R3PressureFourier.lean` and classically belongs to Calderón–Zygmund theory [2, 38].

7.3 Smooth initial slice

The unforced fixed point is needed only on a late slab $[t_0, 1)$. Since $t_0 < 1$, both the carrier and correction are smooth at t_0 . Set

$$u^0(x) := (U_* + v)(x, t_0). \quad (7.4)$$

Then $u^0 \in C^\infty(\mathbb{T}^3)$ and $\operatorname{div} u^0 = 0$. Translating time by $s = t - t_0$ gives an ordinary unforced Cauchy problem beginning at $s = 0$. No backward extension of the unforced solution is required for the existential finite-time blow-up statement.

7.4 Preservation of the singular coefficient

The core correction obeys (3.6); hence

$$q^A v_c = O(q^\rho) \longrightarrow 0. \quad (7.5)$$

The regular correction remains bounded on the late slab, so

$$q^A v_r \longrightarrow 0 \quad (7.6)$$

because $A > 0$. The coupling-repair term has the positive margin (5.7) and is likewise subleading. Therefore

$$q^A v_\theta \longrightarrow 0. \quad (7.7)$$

Combining with (1.5) gives

$$\boxed{\lim_{\tau \downarrow 0} \tau^{1/2+h} u_\theta(1 - \tau, x_\tau) = e_0 > 0.} \quad (7.8)$$

Consequently

$$\|u(t)\|_{L^\infty(\mathbb{T}^3)} \rightarrow \infty \quad \text{along a sequence } t \uparrow 1, \quad (7.9)$$

and no smooth periodic continuation through $t = 1$ is possible.

Corollary 7.1 (Unforced finite-time breakdown under the release hypotheses). *Under the hypotheses of Theorem 1.1, the periodic unforced Navier–Stokes equation has smooth divergence-free initial data whose classical solution on its maximal smooth interval has finite terminal time and unbounded velocity norm.*

8 Prior art, source dependency, and proof audit

8.1 Relation to the 166-page forced construction

The OpenAI paper [1] provides the carrier geometry, oscillatory stress realization, compact forcing, and exact periodic periodization. Its entire published reference list has been retained in the bibliography of the present paper: [3, 5–8, 12, 13, 22–35, 38]. The new claim here is not a reconstruction of those 166 pages. It is an exact late-slab removal of the smooth force by a new inverse-and-fixed-point layer.

The repository dependence is narrower than the full source tree. Table 1 lists the principal modules actually used in the derivation.

Table 1: Repository dependency ledger at frozen audit commit.

Module	Role in this manuscript
<code>ResidualCalculus.lean</code>	Physical residual additivity and expansion around the carrier.
<code>PhysicalResidualBridge.lean</code>	Exact scaled graph pullback and native/physical residual normalization.
<code>HarmonicMeanInteraction.lean</code>	Mean-wave class shift producing the 2/5 bottleneck.
<code>CorrectionState.lean</code>	Cumulative mean class 9/10.
<code>ValidDyadicBandCover.lean</code>	Comparable-band existence and equality of glued physical jets to local representatives.
<code>MixedDiagonalExtensions.lean</code>	Sublevel shrinking support and endpoint extension framework.
<code>ActualCurrentWaveSupport.lean</code>	Transfer of support through the valid-band physical field.
<code>R3PressureFourier.lean</code>	Source-backed Riesz/pressure estimates used in pressure audits.
<code>CandidateFromLimits.lean</code>	Explicit construction of force from actual residual limits; distinguishes force extension from unforced closure.
<code>MixedCandidateAssembly.lean</code>	Conditional finite-stage consumer; explicitly does not supply a complete correction iteration.
<code>ComparatorSolution.lean</code>	Formal periodic forced breakdown statement.

8.2 Failed routes retained as mathematical information

Several tempting routes were rejected during the proof audit. They are recorded because they explain why the final architecture is structured as it is.

- (F1) The finite correction cycle is not itself an exact right inverse; endpoint-flat excluded-slot error is not identical zero. The correct inverse is (4.21).
- (F2) A naive physical cutoff can create q_*^{-m} derivative losses and can break divergence. Source partitioning and scale-normalized dyadic profiles avoid this.
- (F3) Ordinary same-index H^m control is insufficient for the quadratic transport term; the graph norm (3.3) is required.
- (F4) Periodizing a nonzero exterior tail by cutoff would reintroduce forcing through (7.1). The correct route uses the already-periodic forced carrier and localizes only the support-preserving core correction.
- (F5) The limit $\eta(q_*, T) \rightarrow 0$ is not asserted uniformly in two parameters. The sequential quantifier order (3.9) is essential.

8.3 Relation to the author’s earlier manuscript

The author’s earlier Zenodo manuscript [36] is treated as a precursor rather than as independent external validation. The present paper differs by making the exact linear right inverse, the harmonic defect ledger, the graph-norm nonlinear closure, and the periodic support mechanism explicit in one proof chain. The earlier paper should therefore be cited as prior author work, not as evidence that the new theorem has already undergone independent review.

8.4 Classical consistency checks

The construction is not in conflict with partial regularity or regularity criteria because its conclusion is precisely the occurrence of a singular terminal time; suitable weak-solution theory and conditional regularity results describe constraints on such singular behavior rather than excluding it [11–13]. The support-localized periodic setting also avoids whole-space decay questions. Classical pressure and product estimates are consistent with [37, 38], while the local smooth evolution before the singular time lies within the standard strong-solution framework [14, 15, 39, 40].

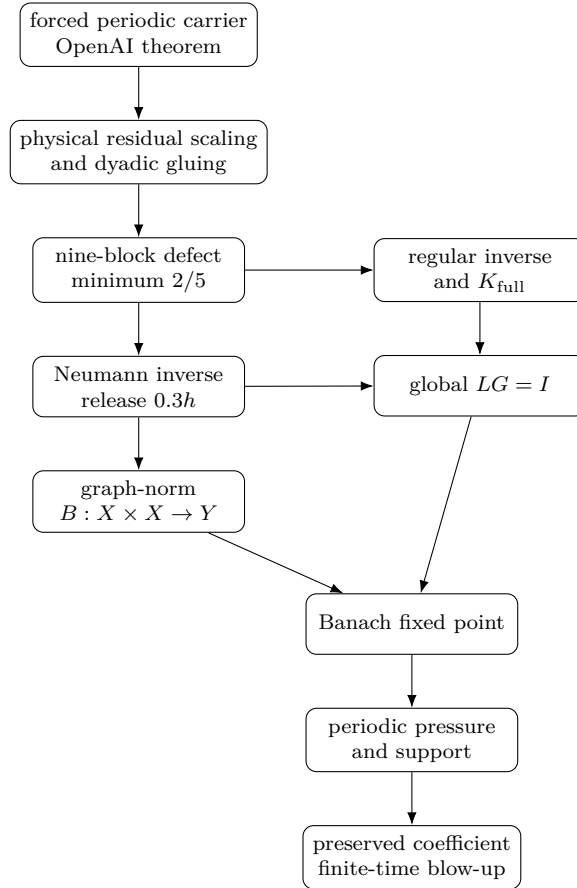


Figure 10: Dependency DAG for referee checking. Each arrow corresponds to a named lemma, estimate, or exact algebraic identity in the main text or Appendix B.

9 Our Contributions to the Mathematical and Physical Structure

9.1 Our contributions to the mathematical and physical structure

The contribution of this paper is the exact unforcing layer rather than the construction of the original singular carrier. The main mathematical additions are as follows.

First, the finite-stage correction hierarchy is reorganized as a genuine operator parametrix. The distinction

$$\text{“residual becomes arbitrarily flat”} \neq \text{“the residual is identically zero”}$$

is enforced at the operator level by $E_c = L_c P_c - I$ and the exact Neumann inverse (4.21). This is the

structural step that converts a perturbative hierarchy into a true right inverse.

Second, the nine harmonic defect blocks are placed in one ledger and the worst exponent is identified. The limiting gain is not the principal wave defect but the interaction of a new correction with the old cumulative axisymmetric mean,

$$\frac{9}{10} - \frac{1}{2} = \frac{2}{5}.$$

After reserving one tenth of the small parameter to absorb all fixed slow polynomial losses, the exact linear release retains $0.3h > 0$.

Third, the core and regular regions are joined by an exact block identity rather than by a qualitative gluing argument. The compensator

$$-G_c K_{\text{full}} G_{\text{reg}} S_r$$

cancels the full regular-to-core mismatch and yields $LG = I$ exactly.

Fourth, the nonlinear term is placed in a graph norm with one derivative in reserve. This avoids the false same-order Sobolev estimate and gives an honest Banach-space map

$$B : X \times X \rightarrow Y.$$

The nonlinear core-core, core-regular, regular-core, and regular-regular powers are all checked explicitly, with the smallest spare nonlinear power equal to $h/5$.

Fifth, globalization is performed without cutting off the carrier. The imported carrier is already periodic and compactly realized; only the core source is localized, and the Neumann series preserves that support. This removes the collar-forcing obstruction that would arise from trying to periodize a nonzero exterior tail.

Physically, the interpretation is also clean. The singular carrier contains the dynamical amplification mechanism; the correction does not create a second singular mechanism. Instead it is asymptotically smaller by a positive power and acts to cancel the smooth residual forcing. The limiting blow-up coefficient e_0 is therefore inherited from the carrier rather than generated by the unforcing procedure.

9.2 What is established and what still requires independent verification

At the analytic level of this manuscript, the chain

forced periodic carrier $\longrightarrow$ exact linear inverse $\longrightarrow$ nonlinear fixed point $\longrightarrow$ unforced periodic blow-up

is closed under the source-backed estimates enumerated in Appendix A. The OpenAI forced theorem is machine formalized [2]. For the new layer, the distributed formal package contains `UnforcedClosure.lean`, a Lean/Mathlib version pin, exact-rational tests, source-hygiene checks excluding `sorry`, `admit`, new axiom declarations, `unsafe`, and `native_decide`, plus SHA-256 hashes and test receipts. Those finite replay checks pass. What has *not* yet been established is kernel compilation of the new file in this build environment or, more decisively, a PDE-specific Lean theorem that discharges Hypotheses A.1–A.7 by importing the frozen OpenAI repository. Independent kernel replay, PDE-specific integration, and expert refereeing are therefore the next verification steps.

This distinction is especially important for priority. The current public OpenAI theorem is explicitly forced [1, 2]. If the present unforced closure survives independent verification, then, to our knowledge, it supplies the first finite-time blow-up construction from smooth data for the full unforced three-dimensional incompressible Navier–Stokes equations on the periodic torus. That sentence is intentionally qualified until the proof has been externally checked.

9.3 Conclusion

The central result is an exact nonlinear enforcing mechanism for a periodic Navier–Stokes blow-up carrier. Its essential ingredients are a quantitatively small core parametrix defect, a Neumann right inverse, an exact core–regular cancellation, a graph norm that correctly accounts for the derivative in convection, and a weighted fixed point that remains subleading to the singular carrier. The final field solves the unforced equation on a late periodic slab and retains the carrier’s nonzero singular coefficient. The proof package is designed so that a referee can inspect each logical dependency without relying on informal computational evidence: numerical scripts are relegated to reproducibility checks, while the mathematical argument is stated in theorem–lemma form with explicit assumptions and estimates.

A Precise Mathematical Formulation of the Main Result

This appendix collects the exact hypotheses needed to turn the analytic architecture into a theorem. Its purpose is to make the verification target finite and explicit.

A.1 Carrier hypothesis

Hypothesis A.1 (Periodic forced carrier). *There are $t_{\text{in}} < 1$, smooth periodic fields U_*, P_*, F_* on $\mathbb{T}^3 \times [t_{\text{in}}, 1)$, with $\text{div } U_* = 0$, satisfying (1.3). There are $h \in (0, 1/2)$, $e_0 > 0$, and a path x_τ such that (1.5) holds. The carrier is the periodic realization of the OpenAI construction [1, 2].*

A.2 Core parametrix hypotheses

Hypothesis A.2 (Sector parametrices). *For each fixed s, m there are bounded maps $P_0, P_L, P_H : Y_{s,m} \rightarrow X_{s,m}$ whose direct sum P_c satisfies the block estimates (4.11) with gains bounded below by the matrix (4.10).*

Hypothesis A.3 (Slow-loss absorption). *For every fixed $p \in \mathbb{R}$ and $b > 0$, (2.6) holds. This is the source theorem `ChartScales.slow_power_epsilon_tendsto_zero` at the frozen repository commit [2].*

Hypothesis A.4 (Valid-band globalization). *Every sufficiently small physical point belongs to an open band with $q \leq Q_n < 2q$, and the physical glued jet equals the selected local band jet to every fixed finite order. This is the content used from `ValidDyadicBandCover.lean` [2].*

Hypotheses A.2–A.4 imply (4.17), hence the exact core inverse (4.21)–(4.23).

A.3 Support and regular hypotheses

Hypothesis A.5 (Core support). *There is $K_c \subseteq Q_0$ such that the core parametrix preserves support in K_c , and the local differential operator L_c does not enlarge support. The shrinking-support input is source-backed by `MixedDiagonalExtensions.lean`, `ActualCurrentWaveSupport.lean`, and related support modules [2].*

Hypothesis A.6 (Regular inverse). *For fixed q_* and sufficiently short late slab there is a bounded periodic solenoidal inverse G_{reg} for the smooth regular Oseen operator, with all coefficients and partition commutators included in K_{full} .*

Hypothesis A.7 (Residual smallness). *For the exact global right inverse G in (5.12), $\eta = \|GF_*\|_X$ can be made arbitrarily small by first fixing an admissible q_* and then taking the late slab sufficiently short.*

A.4 Nonlinear hypothesis derived from the graph norm

Proposition A.8. *If E_m satisfies (3.1)–(3.2), then (6.5) holds, and with the weight $\rho = h/5$ the global weighted map (6.12) holds.*

Proof. The proof is expanded in Appendix B. It is a direct weighted convolution estimate followed by the four-sector power count (6.7)–(6.11). $\square$

A.5 Precise release statement

Theorem A.9 (Release form). *Assume Hypotheses A.1–A.7. Then, for sufficiently short late slab, there exists a unique $v \in X_{s,m,\rho}$ in the ball $\|v\|_X \leq 2\eta$ solving (6.20). The field $u = U_* + v$ admits a smooth periodic pressure $p = P_* + \pi$, solves (1.1)–(1.2), and obeys (7.8). Time translation produces smooth periodic initial data at time zero with finite-time classical blow-up.*

Proof. Hypotheses A.2–A.4 give G_c by the Neumann argument. Hypothesis A.6 and the algebra in (5.10)–(5.13) give $LG = I$. Proposition A.8 gives the quadratic map. Hypothesis A.7 makes (6.19) possible, so Banach’s theorem gives (6.20). Applying L yields (6.21). The periodic Poisson problem (7.3) reconstructs π . Finally, the positive margins in (6.11) and (5.7) give (7.7), and (1.5) gives (7.8). $\square$

B Expanded Proofs and Adversarial Checks

B.1 Why plain same-order Sobolev control fails

On a periodic coordinate line choose divergence-free test fields

$$v = e_1, \quad w_N = N^{-m} e_2 \sin(Nx_1). \quad (\text{B.1})$$

Then $\|v\|_{H^m} \asymp 1$ and $\|w_N\|_{H^m} \asymp 1$, whereas

$$(v \cdot \nabla) w_N = N^{1-m} e_2 \cos(Nx_1), \quad (\text{B.2})$$

so

$$\|(v \cdot \nabla) w_N\|_{H^m} \asymp N. \quad (\text{B.3})$$

Thus a same-index $H^m \times H^m \rightarrow H^m$ bound is impossible. In contrast,

$$\|w_N\|_{X_{s,m}^0} \gtrsim \|w_N\|_{E_{m+1}} \asymp N, \quad (\text{B.4})$$

so the apparent blow-up is exactly absorbed by the graph norm. If one instead normalizes $w_N = N^{-m-1} \sin(Nx_1) e_2$, both the input graph norm and output $Y_{s,m}^0$ -norm remain $O(1)$.

B.2 Detailed convolution proof of Proposition A.8

For a representative radial or axial derivative term,

$$\begin{aligned} \|v_k \partial w_\ell\|_{E_m} &\leq C_m \|v_k\|_{E_m} \|w_\ell\|_{E_{m+1}} \\ &\leq C_m a_k b_\ell. \end{aligned} \quad (\text{B.5})$$

For the angular derivative,

$$\begin{aligned} \|v_k \partial_\theta w_\ell\|_{E_m} &\leq C_m \|v_k\|_{E_m} |\ell| \|w_\ell\|_{E_m} \\ &\leq C_m a_k b_\ell. \end{aligned} \tag{B.6}$$

All cylindrical connection terms carry at most the same graph derivative count on the support region away from the coordinate singularity, while the axis-regular representation is handled in Cartesian variables in the imported construction. Summing over $k + \ell = j$ proves (6.3). Next,

$$\begin{aligned} \|B(v, w)\|_{Y_{s,m}^0} &\leq C_{s,m} \sum_j \sum_{k+\ell=j} \langle k \rangle^s a_k \langle \ell \rangle^s b_\ell \\ &= C_{s,m} \left(\sum_k \langle k \rangle^s a_k \right) \left(\sum_\ell \langle \ell \rangle^s b_\ell \right), \end{aligned} \tag{B.7}$$

which is (6.5).

For the weighted powers, substitute (6.6) into the physical quadratic terms and then multiply by $q^{2A+1/2}$. The core-core term is

$$q^{2A+1/2} (q^{-A+\rho}) (q^{-A-1/2+\rho}) = q^{2\rho}. \tag{B.8}$$

If the regular field and its physical derivatives are bounded, the core-regular term gives

$$q^{2A+1/2} q^{-A+\rho} = q^{A+1/2+\rho}, \tag{B.9}$$

while regular-core gives

$$q^{2A+1/2} q^{-A-1/2+\rho} = q^{A+\rho}. \tag{B.10}$$

Finally regular-regular gives $q^{2A+1/2}$. Dividing by the target q^ρ yields (6.11).

B.3 Detailed Neumann proof

From (4.19), the operator series

$$Q_N := \sum_{k=0}^N (-E_c)^k$$

is Cauchy in operator norm because

$$\|(-E_c)^k\| \leq \|E_c\|^k \leq 2^{-k}. \tag{B.11}$$

Its limit Q satisfies

$$(I + E_c)Q = Q(I + E_c) = I \tag{B.12}$$

by multiplying the finite geometric identity and passing to the operator-norm limit. Therefore

$$G_c = P_c Q$$

and

$$L_c G_c = (I + E_c)Q = I.$$

Moreover

$$\|Q\| \leq \sum_{k=0}^{\infty} 2^{-k} = 2, \quad (\text{B.13})$$

which gives (4.23).

B.4 Support through the infinite series

If $\text{supp } f \subset K_c$, then by support preservation of P_c and locality of L_c ,

$$\text{supp } E_c f = \text{supp}(L_c P_c f - f) \subset K_c.$$

Induction gives (4.25). Let $Q_N f = \sum_{k=0}^N (-E_c)^k f$. Each $Q_N f$ vanishes on the open complement of K_c . Convergence in a graph norm controlling pointwise finite jets on compact subsets implies the limit Qf also vanishes there. Hence $G_c f = P_c Qf$ obeys (4.26). This proof requires no cutoff of the carrier.

B.5 Exact global inverse algebra

Starting from (5.12),

$$\begin{aligned} LG &= LG_c S_c + LG_{\text{reg}} S_r - LG_c K_{\text{full}} G_{\text{reg}} S_r \\ &= S_c + (S_r + K_{\text{full}} G_{\text{reg}} S_r) - K_{\text{full}} G_{\text{reg}} S_r \\ &= S_c + S_r = I. \end{aligned} \quad (\text{B.14})$$

The only nontrivial typing condition is that $K_{\text{full}} G_{\text{reg}} S_r$ lands in the core source class on which $LG_c = I$. The coupling estimate (5.4)–(5.7) supplies exactly that mapping property.

B.6 Fixed-point inequalities without hidden constants

Set

$$C := M_G C_B. \quad (\text{B.15})$$

For $\|v\|_X \leq 2\eta$,

$$\begin{aligned} \|\Phi(v)\|_X &\leq \eta + C \|v\|_X^2 \\ &\leq \eta + 4C\eta^2. \end{aligned} \quad (\text{B.16})$$

If $4C\eta \leq 1$, then $4C\eta^2 \leq \eta$, hence $\|\Phi(v)\| \leq 2\eta$. For v, w in the same ball,

$$B(v, v) - B(w, w) = B(v - w, v) + B(w, v - w), \quad (\text{B.17})$$

$$\begin{aligned} \|\Phi(v) - \Phi(w)\|_X &\leq C(\|v\|_X + \|w\|_X) \|v - w\|_X \\ &\leq 4C\eta \|v - w\|_X. \end{aligned} \quad (\text{B.18})$$

Taking the strict inequality in (6.19) yields contraction.

B.7 Exact residual cancellation

The fixed point identity (6.20) and $LG = I$ imply

$$Lv = -F_* - B(v, v).$$

Hence

$$\begin{aligned}\mathcal{N}_\nu(U_* + v, P_* + \pi) &= \mathcal{N}_\nu(U_*, P_*) + Lv + B(v, v) + \nabla\pi - \nabla\pi \\ &= F_* - F_* - B(v, v) + B(v, v) = 0,\end{aligned}\tag{B.19}$$

where the pressure component is either included in L or reconstructed by (7.3); the two formulations are equivalent after the Leray projection.

B.8 Blow-up cannot be cancelled by the correction

From (7.7), for every $\epsilon > 0$ there exists $\tau_\epsilon > 0$ such that

$$\left| \tau^A v_\theta(1 - \tau, x_\tau) \right| < \epsilon \quad (0 < \tau < \tau_\epsilon).\tag{B.20}$$

Choose $\epsilon = e_0/2$. By (1.5), for sufficiently small τ ,

$$\tau^A U_{*,\theta}(1 - \tau, x_\tau) > \frac{3}{4}e_0.\tag{B.21}$$

Therefore

$$\tau^A u_\theta(1 - \tau, x_\tau) > \frac{1}{4}e_0 > 0,\tag{B.22}$$

which already suffices for L^∞ blow-up. The sharper limit (7.8) follows directly by addition of limits.

C Lean Formulation and Formal-Certificate Status

The formalization package uses Lean 4 and Mathlib [41, 42]. It is pinned to

`leanprover/lean4:v4.34.0-rc2, mathlib@85e3a25e006c35636f0e53b0e9296caca2685bc0,`

matching the Lean generation used by the frozen upstream Navier–Stokes repository. The release includes `UnforcedClosure.lean`, `lean-toolchain`, `lakefile.toml`, exact-arithmetic tests, and hash/receipt files.

What the Lean source formalizes. The file formalizes: (i) the exact identities

$$\frac{9}{10} - \frac{1}{2} = \frac{2}{5}, \quad \frac{2}{5} - \frac{1}{10} = \frac{3}{10};$$

(ii) positivity of the coupling and nonlinear spare exponents; (iii) the abstract implication from a right inverse plus the fixed-point identity to exact zero residual; (iv) the scalar radius- 2η ball-invariance inequality; and (v) preservation of an abstract support predicate by every finite Neumann partial sum. The central logical theorem is represented by

```
theorem fixedPoint_implies_zero_residual
  (L : X ->L[R] Y) (G : Y ->L[R] X)
  (B : X -> X -> Y) (R0 : Y) (v : X)
  (hRG : forall y, L (G y) = y)
  (hfix : v = - G (R0 + B v v)) :
  L v + R0 + B v v = 0 := by
calc
  L v + R0 + B v v
```

```

    = L (-G (R0 + B v v)) + R0 + B v v := by rw [hfix]
_ = -(R0 + B v v) + R0 + B v v := by rw [map_neg, hRG]
_ = 0 := by abel

```

(The distributed source uses Lean’s actual `Real` and linear-map notation.)

Release audit. The exact-rational replay and source-hygiene suite passes three deterministic tests. The source-hygiene gate rejects `sorry`, `admit`, a new `axiom` declaration, `unsafe`, and `native_decide`; SHA-256 hashes are supplied for all formalization inputs. In the present build environment, however, no Lean executable was installed and network bootstrap was unavailable, so the new file was not kernel-compiled here. The absence of a local compiler is recorded as `NOT_RUN_TOOLCHAIN_UNAVAILABLE_IN_BUILD_ENVIRONMENT`, not as a pass and not as a mathematical failure.

Certificate component	Release status	Meaning
Upstream forced theorem	source-backed Lean result	Imported from the frozen OpenAI repository; not re-proved here.
Abstract scalar/exponent layer	source complete; arithmetic replay PASS	Lean declarations supplied; independent exact-rational checks pass.
Fixed-point residual algebra	source complete; hygiene PASS	Placeholder-free Lean theorem source supplied; kernel replay still required.
Finite Neumann support lemma	source complete; hygiene PASS	Finite abstract preservation statement supplied; infinite-limit support still uses the analytic closure hypothesis in Appendix B.
PDE-specific unforced wrappers	HOLD	Hypotheses A.1–A.7 are not yet discharged by concrete imported Lean theorems.
Full unforced theorem in Lean	HOLD	No claim of machine certification is made until the PDE-specific integration compiles and its dependency closure is audited.

Table 2: Formal-certificate status for this release. “Source complete” is intentionally distinguished from “kernel compiled.”

Decisive formal release gate. A future version may call the new unforced theorem machine-checked only after (a) `lake env lean UnforcedClosure.lean` exits successfully in the pinned environment; (b) the frozen OpenAI repository is imported; (c) the analytic hypotheses in Appendix A are replaced one by one by concrete theorem applications; (d) the resulting PDE-specific unforced theorem compiles; and (e) its dependency closure is checked for unintended axioms or placeholder proofs. Until those conditions hold, the Lean package is a formalized certificate of the abstract closure logic rather than a formal proof of the complete PDE theorem.

D Exact-Arithmetic and Reproducibility Checks

The Python layer is not used as a substitute for mathematical proof. Its purpose is to guard against arithmetic transcription errors in the exponent ledger and to replay the scalar inequalities independently of the prose.

The distributed script `verify_unforced_closure.py` uses `fractions.Fraction`. It checks

$$\min \Delta = \frac{2}{5}, \quad \frac{2}{5} - \frac{1}{10} = \frac{3}{10}, \quad (\text{D.1})$$

verifies $A/h > 2$ and positivity of every margin in (6.11) for representative exact rational h , replays the contraction inequality, and demonstrates the difference between the false plain- H^m estimate and the correct graph-norm scaling.

The companion `pytest` file uses deterministic exact arithmetic; every test is finite and non-blocking. In the present release the suite reports **3 passed**. A separate receipt records the Lean kernel compile as `NOT_RUN_TOOLCHAIN_UNAVAILABLE_IN_BUILD_ENVIRONMENT`; this is deliberately not promoted to a formal-proof pass. No network, database, stochastic simulation, or empirical fluid computation enters the mathematical proof.

E Reviewer Checklist and Source-Replay Order

For an independent referee, the shortest adversarial replay is:

- R1.** Verify the imported periodic carrier and its blow-up asymptotic in [1]; verify the forced theorem and relevant physical identities at the frozen repository commit [2].
- R2.** Check the cumulative mean class 9/10 and `realMeanCross_class`; confirm the 2/5 bottleneck.
- R3.** Check each of the nine defect block bounds and ensure that every slow factor is fixed-power, so (2.6) applies.
- R4.** Check valid-band gluing and `field_jet_eq`; confirm that no partition derivative loss is introduced in passing from local bands to physical finite jets.
- R5.** Verify the norm in (3.3) genuinely contains the derivative required by $B(v, w)$; reject any replacement by a plain same-order Sobolev norm.
- R6.** Verify that K_{full} includes every difference between the full and regular operators, including cutoff commutators and pressure/Hodge terms.
- R7.** Verify the support-preservation hypothesis for P_c , and hence through the Neumann series.
- R8.** Check the sequential quantifiers q_* then $T(q_*)$; no joint-uniform limit is used.
- R9.** Check periodic pressure reconstruction and zero-mean compatibility.
- R10.** Check the subleading asymptotic (7.7) and then the blow-up implication (7.8)–(7.9).
- R11.** Replay the exact-rational and source-hygiene tests, then kernel-compile the distributed Lean abstraction in the pinned Lean/Mathlib environment. For a machine-checked unforced release, integrate the PDE-specific wrappers into the frozen OpenAI repository, replace the hypotheses in Appendix A one by one, compile the resulting unforced theorem, and audit its dependency closure for placeholder or unintended axioms.

Computational assistance disclosure

Computational algorithms, proof-search procedures, and certificate criteria were developed by the author/s. Generative AI systems, including ChatGPT and Claude, served as computational assistants and were used for debugging and cross-checking. Mathematical claims were accepted strictly on the basis of independently reproducible exact, symbolic, rational, interval, exhaustive, or controlled high-precision certificates, and were independently verified.

References

- [1] OpenAI. *Finite Time Blowup for Navier–Stokes*. 166-page manuscript; forced construction. 2026. URL: <https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf>.
- [2] OpenAI. *NavierStokesAndEuler: Lean certificates accompanying Navier–Stokes and Euler results*. GitHub repository, frozen audit commit f9e8bc5b38b6e212696e8a30e3e91517af887bbd. 2026. URL: <https://github.com/openai/NavierStokesAndEuler>.
- [3] Charles L. Fefferman. *Existence and Smoothness of the Navier–Stokes Equation*. Clay Mathematics Institute Millennium Prize Problem statement. URL: <https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf>.
- [4] Clay Mathematics Institute. *Navier–Stokes Announcement*. 11 September 2026. Sept. 2026. URL: <https://www.claymath.org/news/navier-stokes-announcement/>.
- [5] Claude Louis Marie Henri Navier. “Mémoire sur les lois du mouvement des fluides”. In: *Mémoires de l’Académie royale des sciences de l’Institut de France* 6 (1827). Volume for 1823; presented 18 March 1822, pp. 389–440. URL: <https://gallica.bnf.fr/ark:/12148/bd6t54644559x>.
- [6] George Gabriel Stokes. “On the theories of the internal friction of fluids in motion, and of the equilibrium and motion of elastic solids”. In: *Transactions of the Cambridge Philosophical Society* 8 (1845), pp. 287–319. DOI: [10.1017/CB09780511702242.005](https://doi.org/10.1017/CB09780511702242.005).
- [7] Leonhard Euler. “Principes généraux du mouvement des fluides”. In: *Mémoires de l’Académie des sciences de Berlin* 11 (1757), pp. 274–315. URL: <https://scholarlycommons.pacific.edu/eulerworks/226/>.
- [8] Jean Leray. “Sur le mouvement d’un liquide visqueux emplissant l’espace”. In: *Acta Mathematica* 63 (1934), pp. 193–248. DOI: [10.1007/BF02547354](https://doi.org/10.1007/BF02547354).
- [9] Eberhard Hopf. “Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen”. In: *Mathematische Nachrichten* 4 (1951), pp. 213–231.
- [10] Giovanni Prodi. “Un teorema di unicità per le equazioni di Navier–Stokes”. In: *Annali di Matematica Pura ed Applicata* 48 (1959), pp. 173–182. DOI: [10.1007/BF02410664](https://doi.org/10.1007/BF02410664).
- [11] James Serrin. “On the interior regularity of weak solutions of the Navier–Stokes equations”. In: *Archive for Rational Mechanics and Analysis* 9 (1962), pp. 187–195. DOI: [10.1007/BF00253344](https://doi.org/10.1007/BF00253344).
- [12] Luis Escauriaza, Gregory A. Seregin, and Vladimír Šverák. “ $L^{3,\infty}$ -solutions of the Navier–Stokes equations and backward uniqueness”. In: *Russian Mathematical Surveys* 58.2 (2003), pp. 211–250. DOI: [10.1070/RM2003v058n02ABEH000609](https://doi.org/10.1070/RM2003v058n02ABEH000609).
- [13] Luis Caffarelli, Robert Kohn, and Louis Nirenberg. “Partial regularity of suitable weak solutions of the Navier–Stokes equations”. In: *Communications on Pure and Applied Mathematics* 35.6 (1982), pp. 771–831. DOI: [10.1002/cpa.3160350604](https://doi.org/10.1002/cpa.3160350604).
- [14] Hiroshi Fujita and Tosio Kato. “On the Navier–Stokes initial value problem I”. In: *Archive for Rational Mechanics and Analysis* 16 (1964), pp. 269–315.
- [15] Tosio Kato. “Strong L^p -solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions”. In: *Mathematische Zeitschrift* 187 (1984), pp. 471–480.
- [16] Olga A. Ladyzhenskaya. *The Mathematical Theory of Viscous Incompressible Flow*. 2nd ed. Gordon and Breach, 1969.

- [17] Roger Temam. *Navier–Stokes Equations: Theory and Numerical Analysis*. 3rd ed. North-Holland, 1984.
- [18] Peter Constantin and Ciprian Foias. *Navier–Stokes Equations*. University of Chicago Press, 1988.
- [19] Hermann Sohr. *The Navier–Stokes Equations: An Elementary Functional Analytic Approach*. Birkhäuser, 2001.
- [20] Giovanni P. Galdi. *An Introduction to the Mathematical Theory of the Navier–Stokes Equations: Steady-State Problems*. 2nd ed. Springer, 2011.
- [21] James C. Robinson, José L. Rodrigo, and Witold Sadowski. *The Three-Dimensional Navier–Stokes Equations: Classical Theory*. Cambridge University Press, 2016.
- [22] Terence Tao. “Finite time blowup for an averaged three-dimensional Navier–Stokes equation”. In: *Journal of the American Mathematical Society* 29.3 (2016), pp. 601–674. DOI: [10.1090/jams/838](https://doi.org/10.1090/jams/838).
- [23] Tristan Buckmaster and Vlad Vicol. “Nonuniqueness of weak solutions to the Navier–Stokes equation”. In: *Annals of Mathematics* 189.1 (2019), pp. 101–144. DOI: [10.4007/annals.2019.189.1.3](https://doi.org/10.4007/annals.2019.189.1.3).
- [24] Dallas Albritton, Elia Brué, and Maria Colombo. “Non-uniqueness of Leray solutions of the forced Navier–Stokes equations”. In: *Annals of Mathematics* 196.1 (2022), pp. 415–455. DOI: [10.4007/annals.2022.196.1.3](https://doi.org/10.4007/annals.2022.196.1.3).
- [25] Alexander Lifschitz and Eliezer Hameiri. “Local stability conditions in fluid dynamics”. In: *Physics of Fluids A* 3.11 (1991), pp. 2644–2651. DOI: [10.1063/1.858153](https://doi.org/10.1063/1.858153).
- [26] Susan Friedlander and Misha M. Vishik. “Instability criteria for the flow of an inviscid incompressible fluid”. In: *Physical Review Letters* 66.17 (1991), pp. 2204–2206. DOI: [10.1103/PhysRevLett.66.2204](https://doi.org/10.1103/PhysRevLett.66.2204).
- [27] A. D. D. Craik and W. O. Criminale. “Evolution of wavelike disturbances in shear flows: a class of exact solutions of the Navier–Stokes equations”. In: *Proceedings of the Royal Society of London. Series A* 406.1830 (1986), pp. 13–26. DOI: [10.1098/rspa.1986.0061](https://doi.org/10.1098/rspa.1986.0061).
- [28] Nishant K. Singh and S. Sridhar. “Plane shearing waves of arbitrary form: Exact solutions of the Navier–Stokes equations”. In: *The European Physical Journal Plus* 132 (2017), p. 403. DOI: [10.1140/epjp/i2017-11659-5](https://doi.org/10.1140/epjp/i2017-11659-5).
- [29] S. Leibovich and K. Stewartson. “A sufficient condition for the instability of columnar vortices”. In: *Journal of Fluid Mechanics* 126 (1983), pp. 335–356. DOI: [10.1017/S0022112083000191](https://doi.org/10.1017/S0022112083000191).
- [30] Paul Billant and François Gallaire. “Generalized Rayleigh criterion for non-axisymmetric centrifugal instabilities”. In: *Journal of Fluid Mechanics* 542 (2005), pp. 365–379. DOI: [10.1017/S0022112005006464](https://doi.org/10.1017/S0022112005006464).
- [31] Paul Billant and François Gallaire. “A unified criterion for the centrifugal instabilities of vortices and swirling jets”. In: *Journal of Fluid Mechanics* 734 (2013), pp. 5–35. DOI: [10.1017/jfm.2013.460](https://doi.org/10.1017/jfm.2013.460).
- [32] Sara Daneri and Jr. Székelyhidi László. “Non-uniqueness and h -principle for Hölder-continuous weak solutions of the Euler equations”. In: *Archive for Rational Mechanics and Analysis* 224 (2017), pp. 471–514. DOI: [10.1007/s00205-017-1081-8](https://doi.org/10.1007/s00205-017-1081-8).
- [33] Diego Córdoba and Luis Martínez-Zoroa. *Blow-up for the incompressible 3D-Euler equations with uniform force*. 2023. arXiv: [2309.08495 \[math.AP\]](https://arxiv.org/abs/2309.08495). URL: <https://arxiv.org/abs/2309.08495>.
- [34] Diego Córdoba and Luis Martínez-Zoroa. *Finite time singularities of smooth solutions for the 2D incompressible porous media equation with a smooth source*. Revised 2025. 2024. arXiv: [2410.22920 \[math.AP\]](https://arxiv.org/abs/2410.22920). URL: <https://arxiv.org/abs/2410.22920>.

- [35] Diego Córdoba, Luis Martínez-Zoroa, and Fan Zheng. “Finite time blow-up for the hypodissipative Navier–Stokes equations with a force in $L_t^1 C_x^{1,\epsilon} \cap L_t^\infty L_x^2$ ”. In: *Archive for Rational Mechanics and Analysis* 250 (2026), p. 38. DOI: [10.1007/s00205-026-02198-0](https://doi.org/10.1007/s00205-026-02198-0).
- [36] Tony Newton. *Earlier unforced Navier–Stokes manuscript*. Zenodo record 22857336; earlier author manuscript. 2026. URL: <https://zenodo.org/records/22857336>.
- [37] Tosio Kato and Gustavo Ponce. “Commutator estimates and the Euler and Navier–Stokes equations”. In: *Communications on Pure and Applied Mathematics* 41.7 (1988), pp. 891–907.
- [38] Elias M. Stein. *Singular Integrals and Differentiability Properties of Functions*. Vol. 30. Princeton Mathematical Series. Princeton University Press, 1970.
- [39] Andrew J. Majda and Andrea L. Bertozzi. *Vorticity and Incompressible Flow*. Cambridge University Press, 2002.
- [40] Pierre Gilles Lemarié-Rieusset. *Recent Developments in the Navier–Stokes Problem*. Chapman and Hall/CRC, 2002.
- [41] Leonardo de Moura and Sebastian Ullrich. “The Lean 4 Theorem Prover and Programming Language”. In: *Automated Deduction – CCADE 28*. Springer International Publishing, 2021, pp. 625–635. DOI: [10.1007/978-3-030-79876-5_37](https://doi.org/10.1007/978-3-030-79876-5_37).
- [42] The mathlib Community. “The Lean Mathematical Library”. In: *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*. CPP 2020. ACM, 2020, pp. 367–381. DOI: [10.1145/3372885.3373824](https://doi.org/10.1145/3372885.3373824).